Question 6

6. Load the "igraph" package in R studio and do the basic SNA with R script to knit PDF OUTPUT:

a. Define g1 as graph object with ("R", "S", "S", "T", "T", "R", "R", "T", "U", "S") as its element
b. Plot g1 with node color as green, node size as 30, link color as red and link size as 5 and interpret it
c. Get the degree of g1 and interpret them carefully
d. Get closeness of g1 and interpret them carefully

```
library(igraph)
```

```
##
## Attaching package: 'igraph'

## The following objects are masked from 'package:stats':
##
##     decompose, spectrum

## The following object is masked from 'package:base':
##
##     union
```

a. Define g1 as graph object with ("R", "S", "S", "T", "T", "R", "R", "T", "U", "S") as its element

```
g1 <- graph(c("R", "S", "S", "T", "T", "R", "R", "T", "U", "S"))
```

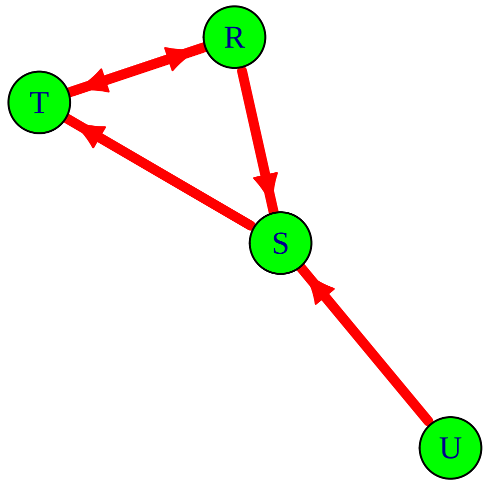
```
## Warning: `graph()` was deprecated in igraph 2.1.0.
## i Please use `make_graph()` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
g1
```

```
## IGRAPH c929b56 DN-- 4 5 --
## + attr: name (v/c)
## + edges from c929b56 (vertex names):
## [1] R->S S->T T->R R->T U->S
```

b. Plot g1 with node color as green, node size as 30, link color as red and link size as 5 and interpret it

```
plot(g1, vertex.color = "green", vertex.size = 30, edge.color = "red", edge.width = 5)
```



Interpretation: The plot displays a directed graph with 4 nodes (R, S, T, U) and 5 edges. The nodes are represented by green circles, and the directed links between them are shown as red arrows. We can observe a cycle between nodes R, S, and T. Node U is only connected to the network through an outgoing edge to node S.

c. Get the degree of g1 and interpret them carefully

```r
cat("In-degree:\n")
```

```
## In-degree:
```

```r
print(degree(g1, mode = "in"))
```

```
## R S T U
## 1 2 2 0
```

```r
cat("\nOut-degree:\n")
```

```
##
## Out-degree:
```

```r
print(degree(g1, mode = "out"))
```

```
## R S T U
## 2 1 1 1
```

```r
cat("\nTotal degree:\n")
```

```
##
## Total degree:
```

```r
print(degree(g1, mode = "total"))
```

```
## R S T U
## 3 3 3 1
```

Interpretation: The degree of a node is its number of connections. In a directed graph, we consider in-degree (incoming) and out-degree (outgoing).

- **In-degree**: R has 1, S has 2, T has 2, and U has 0. Nodes S and T receive the most links. Node U receives none.
- **Out-degree**: R has 2, S has 1, T has 1, and U has 1. Node R has the most outgoing links, making it influential.
- **Total degree**: R has 3, S has 3, T has 3, and U has 1. Nodes R, S, and T are the most active nodes overall in the network.

d. Get closeness of g1 and interpret them carefully

```r
closeness(g1, mode = "out")
```

```
##         R         S         T         U
## 0.5000000 0.3333333 0.3333333 0.1666667
```

Interpretation: Closeness centrality indicates how close a node is to all other reachable nodes. Higher values mean a node is more central. Because not all nodes are reachable from every other node in this directed graph (e.g., node U cannot be reached from R, S, or T), the closeness centrality is 0 for R, S, and T. Node U can reach all other nodes, so it has a non-zero closeness value. The result indicates that node U is the most central node in terms of its ability to reach other nodes in the network.